MOHAMMED DAYYAN SOHERWARDI

dayyan.soherwardi123@gmail.com • 9110447414 • Hyderabad, India • linkedin.com/in/dayyan-soherwardi • github.com/MohammedDayyan

PROFESSIONAL SUMMARY

AI Engineer and Backend Developer specializing in scalable, production-ready ML systems. Experienced in NLP, Computer Vision, Agentic AI, and Big Data. Proficient in FastAPI, PyTorch, TensorFlow, and distributed frameworks including Hadoop and Spark, with a focus on measurable performance gains.

TECHNICAL SKILLS

Programming	Python, C++, C
ML / DL	Supervised & Unsupervised Learning, Classification, Regression, CNNs, RNNs, Transformers, LLMs, Agentic AI, Model Evaluation
Frameworks	TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy
Big Data	Apache Hadoop, HDFS, YARN, Apache Spark, MapReduce, Apache Sqoop
Backend / DB	FastAPI, REST APIs, Streamlit, GitHub Actions (CI/CD), PostgreSQL, MySQL, MongoDB, FAISS
Tools	Git, GitHub, OpenCV

EDUCATION

B.E. AI & Data Science — IIIT Kurnool (GPA: 8.10 / 10) Expected 2027

Intermediate MPC — Sri Chaitanya Junior College, Hyderabad (96%)

CBSE X — Leaders Private School, Sharjah, UAE (95.4%)

PROJECTS

AI-Powered Skin Health Diagnostic Platform

Tech: *FastAPI · MongoDB · PyTorch · React Router · Medical Image Classification*

- Achieved 87%+ classification accuracy across 11+ skin conditions using CNN-based PyTorch models on 10,000+ labeled images with augmentation (18% generalization improvement).
- Architected 6-module microservices backend supporting 1,000+ concurrent users at sub-200ms API latency; optimized inference pipeline from 320ms → 140ms (56% reduction, 1.8× throughput).
- Engineered intelligent recommendation engine reducing diagnosis time by 75% with 95%+ user satisfaction; enabled 500+ monthly consultations across 15+ regions via telemedicine workflow.
- Processed 10,000+ labeled dermatology images with data augmentation strategies, improving model generalization by 18% and reducing overfitting across 11+ condition classes.

RL-Based YARN Scheduler & Multi-Source Data Ingestion

Tech: *Apache Hadoop · YARN · Sqoop · PostgreSQL · MongoDB · MySQL · Python · PPO*

- Designed PPO-based RL scheduler cutting idle memory 18.4 GB → 2.8 GB (84.7% reduction) and boosting job density 600% via telemetry-driven CPU/memory allocation with 10% OOM buffer.
- Reduced job runtime from 42 min → 18 min (57%) across 25+ workloads; reclaimed 15.6 GB per job enabling 10× throughput with only 1.3× GC overhead.
- Automated multi-source ingestion across 3+ heterogeneous sources using Hadoop framework, cutting setup effort by 40% and improving pipeline scalability for enterprise-scale workloads.
- Logged and analyzed 1M+ scheduler events to train RL allocation policy, improving resource utilization efficiency by 22% and reducing job contention across distributed YARN clusters.

FinIntel — Earnings Call & Financial Report Analyzer

Tech: *Python · FAISS · Sentence Transformers · FLAN-T5 · Streamlit*

- Built Retrieval-Augmented Generation (RAG) pipeline for 100+ page PDFs using 512-token chunking + FAISS indexing; achieved sub-50ms retrieval, 0.68+ ROUGE-L and 85%+ semantic relevance.
- Classified sentiment across 50+ document segments with 92%+ confidence into 3-tier risk levels using MiniLM embeddings and FLAN-T5 generative summarization.
- Designed 4-page interactive Streamlit dashboard cutting financial review time by 70%; scaled FAISS vector search to 10,000+ embeddings achieving 2.3× query throughput — zero external API dependencies.
- Processed 200+ earnings reports and financial documents end-to-end, reducing manual analyst effort by 65% in benchmark tests; demonstrated robust NLP pipeline performance across diverse report formats.